

Introduction to Database Systems

The Recipe Hub

Implementation Phase

Team Number: 15

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Problem Definition

The goal of this project is to create a comprehensive recipe database system that can serve a wide spectrum of users; such as cooking enthusiasts, chefs, companies and nutritionists. The database will function as a dynamic platform for finding, exchanging and engaging with recipes that are suited to different cooking methods, dietary requirements and nutritional objectives.

Cooking enthusiasts can explore and save recipes, join community groups and create meal plans. Chefs can share their culinary creations, engage with followers, and highlight special recipes. Companies can promote their products through recipes, gaining insights into consumer preferences, while nutritionists offer guidance on healthy eating by sharing diet plans and nutritional advice. The system's interactive functionalities and user-focused features make it a versatile resource for fostering culinary inspiration, promoting healthy habits, and supporting brand engagement.

System Users

Cooking Enthusiasts

People looking to try new recipes or look to impress their families with a new amazing recipe.

Chefs

Chefs that are willing to share their recipes publicly and inspire other young chefs.

Companies

Companies willing to promote their products by sharing recipes using the product.

Nutritionists

Health experts dedicated to promoting balanced, nutritious diets and guide users with recipes.

Privileges

Admin:

- ★ Accept or decline Chef/nutritionist/company

Cooking Enthusiasts:

- ★ Search for recipes based on filters. (chef/nutritionist/recipe name, time range, difficulty level, top 5)
- ★ Add recipes to favorites
- ★ like recipes.
- ★ Rate and comment on recipes
- ★ Generate shopping lists based on recipe ingredients.
- ★ Create a "To-Cook" List
- ★ Mark Recipes as Tried.
- ★ Follow chefs & nutritionists.

Chefs:

- ★ Add, edit and update recipes.
- ★ Share recipes publicly.
- ★ Respond to comments and questions.
- ★ Partner with food brands / restaurants
- ★ Suggest substitutes for certain ingredients in their recipes. (e.g. homemade or canned salsa)
- ★ Create recipe collections (e.g. holiday specials, seasonal dishes)
- ★ Set difficulty levels for recipes (e.g., easy, intermediate, expert).
- ★ View recipe performance by tracking ratings, reviews,likes, and popularity of each recipe

Companies:

- ★ Create/add recipes that highlight their product. (e.g. Koko creating a recipe using their chicken stripes)
- ★ Delete recipes with products that they no longer sell
- ★ Update product-related recipes as needed.
- ★ Track performance of recipes by seeing engagement, likes count.
- ★ Collaborate with top chefs.

Nutritionists:

- ★ Create/add diet plans and guide users on certain foods based on the user's goal (e.g. weight loss, bulking, etc..).
- ★ Add nutritional information to recipes and ingredients.
- ★ Provide comments on recipes (e.g. healthier options).
- ★ Provide nutritional information of recipes / ingredients (e.g. Xg of pasta contains Y amount of Kcal).
- ★ Offer calorie calculation for meals based on ingredient quantities.
- ★ Educate the community about healthy eating, food allergies, etc.. in the form of blogs.

Database Constraints

- ★ Email should be unique
- ★ Primary Key Constraints
- ★ Foreign Key Constraints
- ★ Not Null Constraints